

Any deviation from the perfect periodic arrangement of atom in a crystalline material is called defect in crystals.

13.18 Engineering Physics

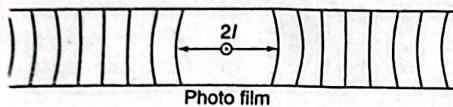


Fig. 13.27

possible diffraction planes will be available for the Bragg diffraction ($2d \sin \theta = n\lambda$) to take place.

All these diffracted rays will lie on a conical surface having its apex at P and semivertical angle 2θ . The diffracted X-ray is recorded by the photographic film placed around the crystal and we get the arc of the circle on the photographic film, as shown in Fig. 13.27.

If $2l$ is the distance of the two arcs of the same circle on the photographic film and R is the radius of cylindrical camera (Fig. 13.28), then

$$2\theta = \frac{l}{R} \quad \text{or} \quad \theta = \frac{l}{2R}$$

This value of θ will give the spacing between the planes with the help of Bragg's relation,

$$2d \sin \theta = n\lambda$$

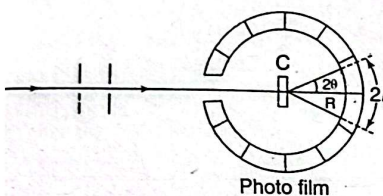


Fig. 13.28

y differentiating

$$2\Delta d \sin \theta + 2d \cos \theta \cdot \Delta \theta = 0$$

$$\text{or} \quad \frac{\Delta \theta}{\Delta d} = -\frac{\tan \theta}{d}$$

θ tends to 90° then $2\theta = 180^\circ$. For this angle, X-rays get reflected back along their initial path and such reflected beams cannot be recorded. For $\theta = 90^\circ$, $(\Delta \theta / \Delta d)$ becomes very large so that small variations in produces large variation in θ .

The most important types of

13.16 POINT DEFECTS IN SOLIDS

Crystal defects: ① Point defects ② line defects ③ surface defects

point defect is a discontinuity in a crystal lattice. It consists of either a missing atom or an ion that creates a vacancy in the lattice (often known as Schottky defect). If an extra atom or ion exists between two normal lattice points it is said to create an interstitial position and if the missing atom or ion shifts to an interstitial position, then the vacancy is called Frenkel defect. Point defect occurs because of the absence of a matrix atom or the presence of an impurity atom or the matrix atom at the wrong place. The most common type point defects in a pure crystal are given below.

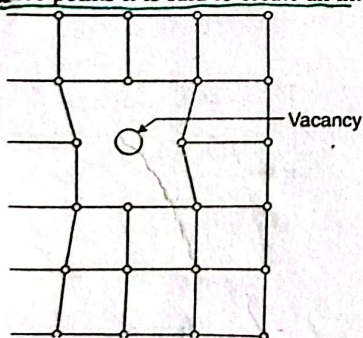


Fig. 13.29

13.16.1 Vacancies

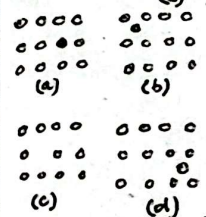
Vacancies are created during crystallisation or from thermal vibrations of the atoms at high temperatures. During thermal vibration, the atoms may acquire sufficiently high energy and evaporate partially or completely and hence create a vacancy (vacancies) in the lattice (Fig. 13.29).

In an ionic crystal, the formation of vacancy requires the charge neutrality which should be maintained in the crystal as a whole. As a

plane defects
① stacking fault
② Grain boundary

Point defects

are of three basic kinds: the substitutional, the interstitial and the vacancy



The defect is due to the presence of regular atom in an interstitial position

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Substitutional: An impurity is substitutional if it occupies a lattice site from which a host atom has been expelled (Fig. 13.29). The impurity is interstitial if it occupies a position between the host atoms. If the impurity atom has roughly the same size and valency as the atoms, then the substitutional impurity is created.

Vacancies: If the atom is missing from its normal site in the matrix, the defect is called a vacancy defect. Vacancies are created in metals, as in other solids, by thermal excitation at high temperatures.

Interstitials: This defect is due to the presence of a regular atom in an interstitial position. A considerable amount of energy is needed to pull out an atom from its regular position and place it in an interstitial position. The defect is created thermally at high temperatures, near the melting point of the solid.

The pair of vacancy and the interstitial atom, thus created, is called Frenkel defect.

Line defects: Line defects in a crystal are called dislocations. A dislocation is a line of atoms with different co-ordinations from those of the other atoms in the crystal.

In plane defects, the crystalline irregularity extends in two dimensions. The atoms at the surface of a crystal are not surrounded by as many numbers of atoms as those inside the crystal.

ies causes to missing of ion from the structure. sites is called a Schottky defect. If a cation goes into an interstitial position, the defect is called a Frenkel defect (Fig. 13.30).

Concentration of defects

Schottky defect, the more than compensated order in the lattice. Due to the disorder increases the entropy. A certain number of vacancies, called the concentration of Schottky defects, are present in thermal equilibrium which can be calculated as follows.

n is the number of Schottky defects produced in removing na^+ cations and na^- anions. N is the total number of cation-anion pairs. If U be the average energy, which is required to create a defect, then nU would be the increase in energy associated with the generation of n defects. The number of different ways in which a cation or an anion can be removed is given by $\frac{N!}{(N-n)!n!}$. If n cations and n anions, the total number of different ways in which n Schottky defects can be created is given by

$$W = \left[\frac{N!}{(N-n)!n!} \right]^2$$

Entropy is given by

$$S = k \ln W$$

$$= k \ln \left[\frac{N!}{(N-n)!n!} \right]^2$$

k is a constant. This increase in entropy produces a change in the Helmholtz free energy F given as

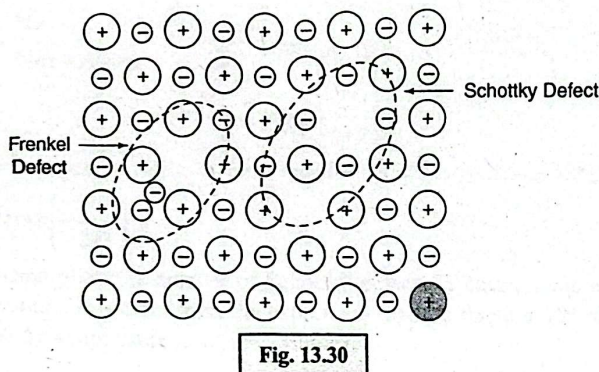
$$= \text{Increase in energy} - \text{Temperature} \times \text{Increase in entropy}$$

$$= nU - 2kT \ln \left[\frac{N!}{(N-n)!n!} \right]$$

Using the approximation $\ln y! = y \ln y - y$, we get the following expression for the change in free energy F

$$= nU - 2kT [N \ln N - (N-n) \ln (N-n) - n \ln n]$$

k is a constant at equilibrium.



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Interstitials: This defect is due to the presence of a regular atom in an interstitial position. A considerable amount of energy is needed to pull out an atom from its